Assignment 4 MATLAB

```
%% Question 1
close all
x=0:0.1*pi:2*pi;
y=sin(x);
plot(x,y)
title('Plot of sinx')
xlabel('x')
ylabel('y')




%% Question 2
close all
x=0:0.1*pi:2*pi;
y1=sin(x);
y2=cos(x);
plot(x,y1,x,y2)
title('Plots')
xlabel('x')
ylabel('y1,y2')

  % Or %

% plot(x,y1)
% hold on
% plot(x,y2)
% title('Plots')
% xlabel('x')
% ylabel('y1,y2')
% hold off

 %(a)

legend('y1','y2');

%(b)

xlim([-1, 2*pi+1])
ylim([-1.5, 1.5])




%% Question 3
close all
x=-1.5:0.1:1.5;

%(a)
```

```
subplot(2,1,1)
y=tan(x);
plot(x,y)

%(b)

title('Graph of tan(x)')
 xlabel('x')
 ylabel('y')

%(c)

subplot(2,1,2)
y=sinh(x);
plot(x,y)

%(d)

title('Graph of sinh(x)')
 xlabel('x')
 ylabel('y')



%% Question 4
close all
% (a)
f1=@(t) 5*t.^2;
fplot(f1)



%% (b)
close
f2=@(t) 5*(sin(t)).^2+(cos(t)).^2;
fplot(f2,[-10 10])



%% (c)
clf
f3=@(t) sin(t)+log(t);
fplot(f3)



%% Question 5

close all
x=0:pi/100:20*pi;
y=x.*sin(x);
z=x.*cos(x);

% (a)
```

```
plot(x,y)
xlabel('x')
ylabel('y')

% % (b)
%
% polarplot(x, y)
%
%
% % (c)
%
% clf
% plot3(x,y,z)
% xlabel('x-axis')
% ylabel('y-axis')
% zlabel('z-axis')
%
%
% % (d)
%
% [X1,Y1]=meshgrid(x,y);
% Z1=X1.*cos(X1);
% contour(Z1)
%
%  % (e)
%
%  surfc(Z1)



%% Question 6

close all
x=-5:0.5:5;
y=-5:0.5:5;
[X,Y] = meshgrid(x,y);
Z=sin(sqrt(X.^2+Y.^2));

% (a)
 mesh(Z)

% % (b)
% subplot(2,1,1)
% surf(Z)
% subplot(2,1,2)
% surf(X,Y,Z)

% % (c)
%
% contour(Z)
%
% % (d)
%
% surfc(Z)
%
```

```
%% Question 7

close all
G = [68, 83, 61, 70, 75, 82, 57, 5, 76, 85, 62, 71, 96, 78, 76, 68, 72, 75, 83, 93];

% (a)

sort(G)
bar(G)
%
% % (b)
%
% hist(G)

%% Question 8

close all
x=linspace(1,5,20);
y=linspace(1,5,20);
[X,Y]=meshgrid(x,y);

% (a)

Z=(X.*exp(X))./(2*Y);
mesh(Z)

% % (b)
%
% Z=log(X.^2+Y.^2);
% mesh(Z)
%
% % (c)
%
% Z=(Y.^2)./(X.^2+Y.^2);
% mesh(Z)
```